GRADE 9 & 10 FOUNDATION HANDBOOK • 10" TABLET REFERENCE EDITION

Ready-Reference Formulas & Exam Speed Codex

Mathematical shortcuts, boundary condition guardrails, invariant equations, and sub-30-second lightning solvers for Foundation, JEE Main, NEET (UG), and National Science Olympiads.

UNIVERSAL META-PROBLEM-SOLVING VERIFICATION FILTERS

Dimensional Homogeneity Filter

Rule: Every additive term and both sides of an equality must possess identical SI dimensions [M^a L^b T^c].

Speed Impact: Instantly eliminates 1 to 2 incorrect multiple-choice options before writing any physics derivation.

Limiting Boundary-Case Stress Test

Rule: Test asymptotic parameter extremes ($\theta \to 0°$, $\theta \to 90°$, $\mu \to 0$, $\mu \to \infty$, $m \to 0$, $M \to \infty$).

Speed Impact: If an acceleration formula on an incline does not yield $a = g \sin\theta$ when $\mu \to 0$, or $a = g$ when $\theta \to 90°$, reject it immediately.

Redox Electron Charge Conservation Balance

Rule: Total ionic charges on the reactant side must strictly match total ionic charges on the product side ($\Sigma q_{initial} = \Sigma q_{final}$).

Speed Impact: Validates redox equation balancing in 5 seconds without re-counting intermediate hydrogen and oxygen atoms.

Stoichiometric Mass & Atom Invariance

Rule: In closed chemical reactions, total grams of reactants equal total grams of products ($\Sigma m_{reactants} = \Sigma m_{products}$).

Speed Impact: Cracks complex multi-stage mixture and gas stoichiometry problems via global mass conservation.

g: 9.8 / 10 m/s^2	**R:** 8.314 J/(mol·K) = 0.0821 L·atm/(mol·K)	**1 F:** 96,500 C/mol e^-	**N_A:** 6.022 × 10^{23} mol^{-1}	**STP V:** 22.7 L (IUPAC) / 22.4 L	**1 atm:** 1.013 bar = 760 mmHg

Galileo's Odd-Integer Free-Fall Law

INVARIANT

$$s_1 : s_2 : s_3 : \dots : s_n = 1 : 3 : 5 : \dots : (2n - 1)$$

Over consecutive equal time intervals Δt from rest under constant acceleration a, distance $s_n = \frac{1}{2} a (\Delta t)^2 (2n - 1)$.

IN-EXAM SPEED HACK
Avoid quadratic displacement equations. If distance in the 1st second is h, distance in the 3rd second is identically 5h.

DOMAIN: Initial velocity must be strictly zero ($u = 0$) and acceleration must be constant.

TRAP: Never apply if body has non-zero initial speed ($u \neq 0$). In that case, $s_n = u \Delta t + \frac{1}{2} a (\Delta t)^2 (2n - 1)$.

DRILL: Free Fall from Tower (NEET (UG) / JEE Main Core) — 10s vs 60s
Distance in 4th second for dropped stone = $(2 \times 4 - 1) \times (\frac{1}{2} g) = 7 \times 4.9 = 34.3$ m directly.

Average Speed for Segmented Journeys

INVARIANT

$$\text{Equal Distances: } v_{avg} = \frac{2 v_1 v_2}{(v_1 + v_2)} \quad | \quad \text{Equal Times: } v_{avg} = \frac{(v_1 + v_2)}{2}$$

Equal distances mandate harmonic mean (time varies inversely with speed); equal times mandate arithmetic mean.

IN-EXAM SPEED HACK
By AM-HM inequality, $v_{avg,\ time} \geq v_{avg,\ dist}$ with equality iff $v_1 = v_2$.

DOMAIN: Applicable strictly when motion occurs along a single straight direction without turning points.

TRAP: Round-trip journeys always have equal distances (use harmonic mean), not equal times.

DRILL: Round-Trip Commute (Grade 9 Foundation / NEET) — 5s vs 35s
Vehicle travels at 40 km/h and returns at 60 km/h: $v_{avg} = 2(40)(60)/(40 + 60) = 48$ km/h (never 50 km/h!).

Stopping Distance & Velocity Scaling Rule

INVARIANT

$$d_{stop} = \frac{v_0^2}{(2a)} \propto v_0^2, \quad t_{stop} = \frac{v_0}{a} \propto v_0$$

From $v^2 = u^2 - 2as$, setting $v = 0$ gives stopping distance directly proportional to the square of initial speed.

IN-EXAM SPEED HACK
If speed is multiplied by factor k, stopping distance multiplies by k^2. Doubling speed (k=2) quadruples stopping distance (4×).

DOMAIN: Braking acceleration a must remain constant throughout retardation.

TRAP: Stopping distance scales as v^2, but stopping time scales linearly as v. Never confuse the two exponents.

DRILL: Automobile Emergency Braking (JEE Main / Foundation) — 5s vs 40s
Car at 50 km/h stops in 12 m. At 150 km/h (3× speed), stopping distance = $12 \times 3^2 = 12 \times 9 = 108$ m instantly.

Freeze-Frame Relative Collision & Overtaking

INVARIANT

$$v_{A/B} = v_A - v_B, \quad a_{A/B} = a_A - a_B, \quad s_{min} = \frac{v_{rel}^2}{(2\,|a_{rel}|)}$$

Transforming into target B's reference frame reduces a two-body coupled differential system into a single 1D stopping calculation.

IN-EXAM SPEED HACK
To prevent rear-end collision when vehicle A approaches vehicle B: minimum safe initial gap is $s_{min} = (v_A - v_B)^2 / (2\, a_{decel})$.

DOMAIN: Relative acceleration a_{rel} must remain constant throughout the braking maneuver.

TRAP: In opposing motion (head-on), remember vector signs: $v_{rel} = v_A - (-v_B) = v_A + v_B$.

DRILL: Highway Relative Tracking (JEE Main 2026 · Shift 2 · Q27) — 15s vs 75s
Cars A (100 km/h) & B (80 km/h): passenger in B observes relative speed $v_{rel} = 100 - 80 = +20$ km/h forward.

The Factored Trajectory Equation

INVARIANT

$$y = x \tan\theta \left(1 - \frac{x}{R}\right)$$

Factoring $x \tan\theta$ from the standard parabola $y = x\tan\theta - [g x^2 / (2 v_0^2 \cos^2\theta)]$ directly exposes horizontal range $R = (v_0^2 \sin 2\theta) / g$.

IN-EXAM SPEED HACK
When given barrier coordinates (x_1, y_1), never expand $\cos^2\theta$. Use $\tan\theta = y_1 / [x_1 (1 - x_1 / R)]$ for an instant 1-line solve.

DOMAIN: Launch and landing points must be at the same vertical datum (level horizontal terrain).

TRAP: Do not use when launching onto an elevated terrace (elevation mismatch breaks level range symmetry).

DRILL: Target Obstacle Clearance (JEE Main Core Kinematics) — 25s vs 110s

Passes through (x=5m, y=1m) with range R=20m: $1 = 5\tan\theta (1 - 5/20) = 5\tan\theta (3/4) \Rightarrow \tan\theta = 4/15$.

Complementary Launch Angles Invariants

INVARIANT

$$R_\theta = R_{90^\circ-\theta} \;|\; R = 4\sqrt{H_1 H_2} \;|\; t_1 t_2 = 2R / g \;|\; H_1 + H_2 = v_0^2 / (2g)$$

Angles θ and $90^\circ - \theta$ yield identical $\sin 2\theta$. Cross products of heights and times cancel trigonometric factors identically.

IN-EXAM SPEED HACK
Given maximum heights H_1 and H_2 of two complementary shots with identical speed, Range is $4\sqrt{H_1 H_2}$ without computing speed v_0 or angle θ.

DOMAIN: Initial speeds must be identical, launched over horizontal ground.

TRAP: Remember factor 4 in $R = 4\sqrt{H_1 H_2}$; forgetting factor 4 is the most common arithmetic trap.

DRILL: Dual Trajectory Heights (NEET (UG) / JEE Main Invariants) — 10s vs 75s

$H_1 = 20$ m, $H_2 = 45$ m. Range $R = 4\sqrt{(20 \times 45)} = 4\sqrt{(900)} = 4 \times 30 = 120$ m directly.

Horizontal Tower Launch & Free-Fall Decoupling

INVARIANT

$$t_{fall} = \sqrt{(2h / g)}, \quad v_x = R / \sqrt{(2h / g)}, \quad v_{impact} = \sqrt{(v_x^2 + 2gh)}$$

Vertical motion is pure free fall from rest independent of horizontal launch speed.

IN-EXAM SPEED HACK
For horizontal launch from height h, horizontal speed is directly $v = R / t = R / \sqrt{(2h / g)}$.

DOMAIN: Initial vertical velocity must be zero (purely horizontal takeoff).

TRAP: Do not treat landing distances as proportional to velocity if vertical fall heights differ.

DRILL: Cliff Projectile Landing (Grade 9 Foundation / NEET) — 15s vs 80s

h = 20m, lands at R = 40m. $t = \sqrt{(2(20)/10)} = 2.0$s $\Rightarrow$ Horizontal launch speed $v_0 = 40 / 2.0 = 20$ m/s.

Maximum Range & Apex Energy Invariants

INVARIANT

$$\text{At } \theta = 45^\circ: R_{max} = v_0^2 / g = 4 H_{max} \;|\; K_{apex} = K_0 \cos^2\theta$$

Maximum range occurs when $\sin 2\theta = 1$ ($\theta = 45^\circ$). Vertical velocity at apex vanishes; horizontal velocity $v_0 \cos\theta$ remains invariant.

IN-EXAM SPEED HACK
At maximum range angle (45°), Range is always exactly 4 times the maximum height ($R = 4H$). Apex kinetic energy is exactly half ($\frac{1}{2} K_0$).

DOMAIN: Constant gravitational acceleration g without atmospheric air drag.

TRAP: Kinetic energy at highest point is not zero; it equals $\frac{1}{2} m (v_0 \cos\theta)^2$, which is non-zero for any oblique angle.

DRILL: Apex Kinetic Energy Fraction (NEET (UG) / Foundation) — 10s vs 50s

If kinetic energy at apex is half of initial KE: $\cos^2\theta = \frac{1}{2} \Rightarrow \cos\theta = 1/\sqrt{2} \Rightarrow \theta = 45^\circ$ (Maximum range angle!).

Whole-System Acceleration Heuristic

INVARIANT

$$a_{sys} = (\Sigma F_{driving} - \Sigma F_{opposing}) / \Sigma m_{inertial}$$

Internal constraint forces (tensions, mutual normal contact forces) form action-reaction pairs that cancel over the total free body.

IN-EXAM SPEED HACK
Never write simultaneous 3-variable equations for Atwood or connected incline systems. One line gives acceleration directly.

DOMAIN: All interconnected components must share identical scalar acceleration magnitude along the constraint path.

TRAP: Include all moving masses in denominator, including pulleys if they have non-negligible rotational inertia.

🎯 DRILL: Standard Atwood Machine (NEET (UG) / Foundation) — 10s vs 60s

Masses 3 kg & 2 kg over smooth pulley: a = (3 - 2)g / (3 + 2) = g / 5 = 2.0 m/s². Tension T = 2(3)(2)g / 5 = 24 N.

Inclined Plane Friction & Time Ratio Scaling

INVARIANT

$$a = g(\sin\theta - \mu\cos\theta) \mid \text{If } t_{rough} = n \cdot t_{smooth} \Rightarrow \mu = \tan\theta\,[1 - (1/n^2)]$$

From $s = \frac{1}{2} a t^2$, acceleration varies inversely as t^2. Thus $a_{rough} / a_{smooth} = 1/n^2$, yielding μ in one line.

IN-EXAM SPEED HACK
If a block takes n times longer to slide down a rough incline than a smooth incline of angle θ, μ = tanθ (1 - 1/n²) directly.

DOMAIN: Incline angle θ must exceed the angle of repose (tanθ > μ) so that motion actually occurs from rest.

TRAP: Factor is (1 - 1/n²), never (1 - 1/n).

🎯 DRILL: Incline Sliding Time Ratio (JEE Main 2026 · Shift 2 · Q46) — 15s vs 90s

Takes twice as long (n=2) on 45° rough plane: μ = tan 45° [1 - 1/(2²)] = 1 × (1 - 1/4) = 0.75 in 5 seconds.

Pulling vs. Pushing Friction Margin

INVARIANT

$$N_{pull} = mg - F\sin\theta \;\; (f_{max} = \mu N_{pull}) \mid N_{push} = mg + F\sin\theta \;\; (f_{max} = \mu N_{push})$$

Pulling has an upward vertical component reducing normal force; pushing has a downward vertical component increasing normal force.

IN-EXAM SPEED HACK
Normal shift is exactly ΔN = 2 F sinθ. Pushing jams completely when tanθ = μ (force required blows up to infinity!).

DOMAIN: Force applied at angle θ to horizontal on rough horizontal surface.

TRAP: If pulling force is very large (F sinθ > mg), the body lifts off the surface (N = 0, friction vanishes completely).

🎯 DRILL: Lawn Roller Dynamics (Grade 9 Foundation / NEET) — 10s vs 50s

Pulling reduces normal force by F sinθ, decreasing friction resistance f = μ N. Pushing adds F sinθ, increasing resistance.

Two-Block Slipping Threshold Rule

INVARIANT

$$a_{max} = \mu_s g, \quad F_{thresh} = (M + m)\,\mu_s g$$

Friction between the blocks is the sole force accelerating the upper block m; its maximum possible acceleration is $\mu_s g$.

IN-EXAM SPEED HACK
Compute system acceleration a_sys = F / (M + m). If a_sys ≤ μ_s g, they move together without slip. If a_sys > μ_s g, slip occurs.

DOMAIN: Smooth horizontal ground under lower block M; friction acts only at mutual interface.

TRAP: Once slipping begins, static friction μ_s must immediately be replaced by kinetic friction μ_k.

🎯 DRILL: Block-on-Block Slip Threshold (JEE Main / Foundation) — 15s vs 75s

m=2kg on M=3kg, μ_s=0.4: a_max = 0.4(10) = 4 m/s². Maximum pulling force without slipping is F_thresh = (2 + 3)(4) = 20 N.

Molarity-to-Molality Direct Conversion

INVARIANT

$$m = 1000\,M / (1000\,d - M\,M_w) \quad | \quad d = (M / 1000)\,[\,(1000 / m) + M_w\,]$$

In 1000 mL solution: total mass = 1000 d grams, solute mass = M M_w grams, solvent mass = (1000 d - M M_w) / 1000 kg.

IN-EXAM SPEED HACK
Eliminates the tedious 2-page 'Assume 1 L solution' algebra. Direct plug-and-chug with density d in g/mL.

DOMAIN: Density d must be in g/mL (or g/cm³). Solute molar mass M_w in g/mol.

TRAP: Ensure d is in g/mL, not kg/m³. If given in kg/m³, divide by 1000 first.

🎯 DRILL: Concentration Unit Interconversion (JEE Main / NEET (UG) Core) — 20s vs 90s

2.0 M aqueous urea (Mw=60) with d=1.06 g/mL: m = 1000(2) / [1000(1.06) - 2(60)] = 2000 / [1060 - 120] = 2000 / 940 = 2.13 m.

Alligation Cross for Binary Mixture Molar Mass

INVARIANT

$$n_1 / n_2 = |M_2 - M_avg| / |M_1 - M_avg|$$

M_avg = ($n_1 M_1 + n_2 M_2$) / ($n_1 + n_2$) rearranged gives $n_1 (M_avg - M_1) = n_2 (M_2 - M_avg)$.

IN-EXAM SPEED HACK
Solves isotope natural abundances, binary alloy compositions, and gas mixture ratios in 5 seconds via cross subtraction.

DOMAIN: Applies strictly to binary (two-component) mixtures.

TRAP: Alligation gives mole ratio for molar properties, but mass ratio for mass-specific properties (like mass percentages).

🎯 DRILL: Partial Pressures in Gas Mixture (NEET (UG) 2020 · Set E1 · Q110) — 15s vs 60s

7g N_2 (0.25 mol) & 8g Ar (0.20 mol): n_ratio = 5:4. Mole fraction of Ar = 4/9. Partial pressure P_Ar = (4/9)(27 bar) = 12 bar.

Faraday's Electrolysis Direct Electron Factor

INVARIANT

$$1\ \text{Faraday} = 96{,}500\ \text{C} \equiv 1\ \text{mole of electrons} \quad | \quad \text{Mass } m = (M / z)\,(Q / F)$$

Reduction of 1 mole of M^(z+) requires exactly z moles of electrons (z Faradays of charge).

IN-EXAM SPEED HACK
Never use complex electrochemical equivalent formulas. Simply use: Moles deposited = Faradays / z.

DOMAIN: 100% current efficiency assumed unless specified otherwise.

TRAP: Identify cation charge z carefully: e.g., Cu^{2+} has z=2, but Cu+ (cuprous) has z=1.

🎯 DRILL: Electrolytic Deposition of Calcium (NEET (UG) 2020 · Set E1 · Q106) — 5s vs 40s

$Ca^{2+} + 2e^- \rightarrow Ca$. Molar mass Ca = 40 g/mol ⇒ 40g requires 2 F. Therefore 20g requires exactly 1.0 F.

Limiting Reagent Dimensionless Ratio Test

INVARIANT

$$R_i = n_i / \nu_i \quad (\nu_i = \text{stoichiometric coefficient})$$

Reaction progress is bounded by the reactant with the minimum value of moles divided by its stoichiometric number.

IN-EXAM SPEED HACK
Divide initial moles of each reactant by its balanced coefficient. Smallest ratio is strictly the Limiting Reagent (LR).

DOMAIN: Reaction must go to completion (irreversible stoichiometry).

TRAP: Never compare raw masses or raw moles directly; always divide by stoichiometric coefficients first.

🎯 DRILL: Haber Ammonia Synthesis (Grade 9 Foundation / NEET) — 10s vs 50s

$N_2 + 3\,H_2 \rightarrow 2\,NH_3$ with 2 mol N_2 and 3 mol H_2. R_N = 2/1 = 2; R_H = 3/3 = 1. H_2 is LR (ratio 1 < 2), yielding 2 mol NH_3.

The 'BAN' Mnemonic for Permanganate ($KMnO_4$)

INVARIANT

Basic (strongly alkaline): n = 1 | Acidic: n = 5 | Neutral / weakly basic: n = 3

Acidic: $MnO_4^- + 5e^- \rightarrow Mn^{2+}$ (+7→+2). Neutral: $MnO_4^- + 3e^- \rightarrow MnO_2$ (+7→+4). Strongly Basic: $MnO_4^- + 1e^- \rightarrow MnO_4^{2-}$ (+7→+6).

IN-EXAM SPEED HACK
Remember keyword 'BAN 1-5-3' (Basic=1, Acidic=5, Neutral=3) to instantly write equivalent weights without balancing half-cells.

DOMAIN: Acidic medium must use dilute H_2SO_4, never concentrated HCl (which oxidizes Cl^- to Cl_2).

TRAP: In neutral or faintly alkaline medium, n = 3 (forms MnO_2), not n = 5.

DRILL: Permanganate Equivalent Weight (NEET (UG) / JEE Main Core) — 5s vs 45s
Equivalent weight of $KMnO_4$ in acid is M/5; in neutral medium (forming MnO_2) it is M/3; in strongly basic medium it is M/1.

Dichromate Acidic Invariant

INVARIANT

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O \quad (n = 2 \times |6 - 3| = 6)$$

Two chromium atoms per formula unit each undergo a 3-electron reduction from +6 to +3.

IN-EXAM SPEED HACK
Equivalent weight of $K_2Cr_2O_7$ in acidic medium is always strictly M / 6.

DOMAIN: Operates as an oxidant in acidic solution; in alkaline medium it hydrolyzes to yellow chromate CrO_4^{2-}.

TRAP: Multiply change in oxidation state per atom (3) by number of Cr atoms in formula unit (2) = 6. Forgetting factor 2 is fatal.

DRILL: Dichromate Charge Transfer (JEE Main / NEET) — 10s vs 45s
Complete reduction of 1 mol $K_2Cr_2O_7$ to Cr^{3+} in acidic medium consumes exactly 6 Faradays (6 × 96,500 C) of charge.

Parallel Resistor Rule for Disproportionation

INVARIANT

$$n_{eq} = (n_{ox} \cdot n_{red}) / (n_{ox} + n_{red}) \iff 1 / n_{eq} = 1 / n_{ox} + 1 / n_{red}$$

Conserving electrons across simultaneously oxidized and reduced branches yields equivalent weight E = M (1/n_ox + 1/n_red).

IN-EXAM SPEED HACK
Identical structure to parallel resistors. Disproportionation n-factor is computed in 5 seconds without writing full balanced stoichiometry.

DOMAIN: Applicable strictly when a single reactant disproportionates into two distinct redox products.

TRAP: Do not simply add n_ox + n_red; addition is only valid when two different elements in the same compound both oxidize.

DRILL: Chlorine Gas Disproportionation (JEE Main / NEET Redox) — 15s vs 70s
$Cl_2 + 2\,OH^- \rightarrow Cl^-$ (n_red = 1) + ClO^- (n_ox = 1). n_eq = (1×1)/(1+1) = 0.5. Equivalent weight of Cl_2 is M / 0.5 = 2M.

The Golden Law of Equivalents (Titration Shortcut)

INVARIANT

$$(N_1\,V_1)_{oxidant} = (N_2\,V_2)_{reductant} \iff n_1 \cdot moles_1 = n_2 \cdot moles_2$$

One equivalent of any oxidizing agent completely consumes exactly one equivalent of any reducing agent.

IN-EXAM SPEED HACK
Never balance full complex redox equations during exam conditions. Multiply moles by respective n-factors and equate directly.

DOMAIN: All reactants must react completely without side reactions.

TRAP: In FeC_2O_4 (ferrous oxalate), both Fe^{2+} ($1e^-$) and $C_2O_4^{2-}$ ($2e^-$) oxidize, so n-factor is 1 + 2 = 3!

DRILL: Oxalic Acid Permanganate Titration (Grade 9-10 Olympiad / NEET) — 15s vs 80s
Oxalic acid ($H_2C_2O_4$, n=2) titrated with acidified $KMnO_4$ (n=5): 2 × moles_ox = 5 × moles_perm ⇒ 5 mol oxalic acid requires 2 mol $KMnO_4$.